INTERACTIVE END-OF-CHAPTER EXERCISES

COMPUTING AN INTERNET CHECKSUM

Consider the two 16-bit words (shown in binary) below. Recall that to compute the Internet checksum of a set of 16-bit words, we compute the one's complement sum [1] of the two words. That is, we add the two numbers together, making sure that any carry into the 17th bit of this initial sum is added back into the 1's place of the resulting sum); we then take the one's complement of the result. Compute the Internet checksum value for these two 16-bit words:

11100000 10011100	*this binary number is 57500 decimal (base 10)*
00000101 11110111	*this binary number is 1527 decimal (base 10)*

QUESTION LIST

1. What is the sum of these two 16 bit numbers? Don't put any spaces in your answer

2. Using the sum from question 1, what is the checksum? Don't put any spaces in your answer

SOLUTION

1. The sum of 11100000 10011100 and 00000101 11110111 = 11100110 10010011

2. The internet checksum is the one's complement of the sum: 11100110 10010011 = 00011001 01101100

QUESTION 1 OF 2

What is the sum of these two 16 bit numbers? Don't put any spaces in your answer

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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